THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 3

Chapter 8: **Nature of Matter: Elements, Compounds and Mixtures** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The components of a mixture can be separated by: 1

(a) no method **(b)** only chemical methods
(c) physical methods **(d)** burning always

2. A molecule of water contains: 1

(a) 1 hydrogen atom and 1 oxygen atom **(b)** 1 hydrogen atom and 2 oxygen atoms
(c) 2 hydrogen atoms and 1 oxygen atom **(d)** 2 hydrogen atoms and 2 oxygen atoms

3. Which of the following is a metalloid? 1

(a) carbon **(b)** iron **(c)** oxygen **(d)** silicon

4. Iron filings can be separated from a mixture with sulphur by using a: 1

(a) magnet **(b)** sieve of any kind
(c) boiling **(d)** filter paper

5. Which of the following is a non-metal that is a good conductor of electricity? 1

(a) phosphorus **(b)** iodine **(c)** sulphur **(d)** graphite

6. Which element is used in making electrical wires? 1

(a) phosphorus **(b)** copper **(c)** iodine **(d)** sulphur

7. When iron and sulphur are heated together, the product (iron sulphide) is a: 1

(a) mixture **(b)** compound **(c)** element **(d)** alloy

8. Which metal is liquid at room temperature? 1

(a) copper **(b)** sodium **(c)** mercury **(d)** iron

9. Gold and silver are used for making jewellery because they are: 1

(a) brittle **(b)** lustrous and malleable
(c) dull **(d)** poor conductors

10. Which of the following is NOT a mixture? 1

(a) tap water **(b)** lemonade **(c)** distilled water **(d)** sea water

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why is sodium stored under kerosene? 2

12. Differentiate between malleability and ductility. 2

13. What is an alloy? Name one. 2

14. Why are the handles of cooking pans made of plastic or wood and not of metal? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Give three examples of mixtures from daily life and name their components. 3

16. How can the components of a mixture of sand, salt and iron filings be separated? Name the steps. 3

17. Write the three states of elements at room temperature with two examples for each. 3

OR

Write the chemical formulae and the elements present in common salt, water and carbon dioxide. 3

18. Explain why the properties of a compound differ from those of its elements with sodium chloride as an example. 3

OR

Why are some elements stored in special ways? Give two examples. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe the uses of metals, non-metals and metalloids with two examples each. 5

OR

Why is air considered a mixture? List its main components with percentages and describe a property of each. 5

20. What is an alloy? Describe how alloys improve properties with examples of steel, brass and bronze. 5

OR

Explain physical and chemical separation using examples such as magnet, filtration, evaporation and electrolysis. 5

END OF QUESTION PAPER — Total: 40 Marks

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